

Requirements Document
for
Orangy Carpels

1. Introduction

Orangy design studio has been in business for 12 years. As the team and project load have grown, managing HR tasks and time tracking through spreadsheets and scattered tools is no longer sustainable.

This document defines the requirements for a custom **Resource, Project & Timesheet Application** — a single platform to handle attendance, employee records, leave, Clients, Project & timesheets management. It is built for internal use, designed around our UX Design studio's workflows, and intended to scale with us long-term.

2. Objectives

- Replace manual HR and timesheet processes with a unified system
- Track employee attendance, leave, and working hours accurately
- Log time spent on projects and tasks for billing and resource planning
- Centralize employee records & documents.
- Give leadership clear visibility into team utilization and project costs

3. Scope

In Scope:

- Employee management (profiles, documents, onboarding)
- Attendance & leave management
- Project-based timesheet tracking
- Role-based dashboards (Admin, Manager, Employee)
- Client management
- Billing Module
- Dashboard & Reports & analytics

Not in Scope (Phase 1):

- Payroll processing support
- Recruitment / hiring module
- Third-party accounting integrations

4. User Roles

Provide a centralized system to manage:

- Employees
- Resource allocation
- Leave management
- Access control
- Roles & hierarchy (CRITICAL)

System must support **strict role-based access control (RBAC)** with the following hierarchy:

4.1 Super Admin

- Full system access
- Can view **financials, costing, billing**
- Final approval authority
- Manages all users and permissions

4.2 Project Manager

- Can:
 - Create/manage projects under assigned clients
 - Assign employees to projects
 - Approve/reject timesheets
 - Manage team (resources)
 - Track hours & utilization
- Cannot:
 - View **costing / billing / financial data**.

4.3 Employee

- Can:
 - Fill timesheets
 - Apply for leave
 - View assigned projects
- Cannot:
 - Approve anything
 - Access financial data

4.4 Resource Management

Features:

- Assign employees to:
 - Projects
- One employee can be:

- Assigned to multiple projects

Project Manager Capabilities:

- Allocate resources
- View team workload
- Track utilization

4.8 Access Control (RBAC)

Rules:

- Super Admin:
 - Full access (including finance)
- Project Manager:
 - No access to:
 - Costing
 - Billing
 - Revenue
- Employees:
 - Limited to:
 - Own data
 - Assigned projects

4.9 Visibility Rules

Feature	Employee	Project Manager	Super Admin
Own Timesheet	✓	✓	✓
Team Timesheets	✗	✓	✓
Project Assignment	✗	✓	✓
Financial Data	✗	✗	✓
Leave Approval	✗	✓	✓

5. Employee Management

- Add, edit, deactivate employee profiles
- Store personal info, designation, department, reporting manager
- Upload documents (ID proof, offer letter, contracts)
- Employee self-service (view & update own details)

Client Fields:

- Full Name
- DOB
- Mobile Number
- Personal Email Address
- Permanent address (Multi Line Value)
- Father's or Guardian name
- Mother's name
- Second phone number
- Blood group
- Educational (Multi Value – User can add “n” number of Educational value)
 - Educational Degree
 - Board or university
 - Institute or college
 - Complete Year
 - Marks or percentage
- Work experience. (Multi Value – User can add “n” number of Work experience value)
 - Employer/Company Name
 - Working from
 - Working To
 - Work Period (months/years)
 - Work City
 - Company reference contact Email
 - Company reference contact Phone Number
- LinkedIn profile link
- AADHAAR number
- PAN number
- Bank Detail
 - Bank name
 - Bank account number
 - Bank Branch
 - IFSC code
 - Attach Passport Size Photograph
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6. Client & Project Management

System must allow creation and management of **Clients & Projects**

Client Fields:

- Client ID (Unique, system-generated or manual)
- Client Name (Company Name)
- Contact Details (Address, Email ID & Phone number)
- Client Billing Currency (USD, INR & All Currency)
- Status (Active / Inactive)

Client Rules:

- Client ID must be unique across the system
- A client can have multiple projects
- Default billing type can be overridden at project level

Project Fields:

- Project ID (Unique, system-generated or manual)
- Project Name
- Billing Type
 1. Hourly rate (T&M),
 2. Monthly Resource Cost (Fixed).
 3. Project cost (Fixed).
- Project Status (Active / Inactive)

7. Attendance & Leave

- Daily check-in / check-out logging
- Leave applies, approve, reject workflow
- Leave balance tracking (casual, sick, earned, comp-off)
- Holiday calendar
- Monthly attendance summary

7.1 Leave Management System

Leave Request Flow

Employee → Project Manager → Super Admin (CEO)

Leave Process

Step 1: Employee

- Apply leave:
 - Date range
 - Leave type
 - Reason

Step 2: Project Manager

- Approve / Reject employee leave
- Manage team availability

Step 3: Super Admin (CEO)

- Approves / reviews **Project Manager leave requests**
- Final authority for escalations

Approval Hierarchy Rules

- Employee leave → approved by Project Manager
- Project Manager leave → approved by Super Admin
- Timesheets:
 - Approved by Project Manager
 - Override by Super Admin

8. Timesheet

- Employee Log hours against clients, projects & tasks
- Allow to Add project cost, AI Token cost & Ext against clients, projects & tasks
- Daily / weekly timesheet entry
- Manager review & approval workflow
- Billable vs non-billable hour tagging

- Team member able to see unlogged time for week & month.

8.1 Timesheet Control (Role-Based)

Project Manager:

- View team timesheets
- Approve / Reject
- Edit (if allowed)

Super Admin:

- Full visibility
- Override approvals

8.2. Timesheet locking & control

Purpose

- Data integrity after submission
- Payroll accuracy
- Prevention of unauthorized changes
- Controlled financial closing process

Locking levels

System must support **multi-level locking**:

8.2.1 Submission Lock

Trigger: When timesheet is submitted for approval

Behavior:

- Employee:
 -  Cannot edit
- Project Manager / Super Admin:
 -  Can edit

8.2.2 Approval Lock

Trigger: When timesheet is approved

Behavior:

- Employee:
 -  No access to edit
- Project Manager:
 -  Limited edit (if allowed by config)
- Super Admin:
 -  Full override

9. Billing Module

- Each project must have only ONE billing type
- 3 Billing Type
 - Hourly rate (T&M),
 - Monthly Resource Cost (Fixed).
 - Project cost (Fixed)
- All Currency values need to convert in to INR finally in dashboard.
- Fetch Currency value live on online API & Weekly One & use for INR conversion
- Rate versioning in history
- Hourly rate (T&M)
 - Hours logged × Hourly Cost
 - Rate versioning from Any currency to INR
 - Multi-currency
 - Project Fixed Cost Fields:
 - Hourly Cost
 - Currency
 - Effective Start Date
 - Effective End Date (optional)
- Resource Price
 - Monthly Resource Cost (Recurring)
 - Fixed amount billed every month
 - Billing NOT dependent on hours, but still timesheet required
 - Multi-currency
 - Project Fixed Cost Fields:
 - Monthly Cost
 - Currency
 - Effective Start Date
 - Effective End Date (optional)
- Project Cost

- Project Cost (one-time or milestone-based – future scope)
- Billing NOT dependent on hours, but still timesheet required
- Multi-currency
- Project Fixed Cost Fields:
 - Monthly Cost
 - Currency
 - Effective Start Date
 - Effective End Date (optional)
- Rate versioning Hourly rate (T&M), Monthly Resource Cost (Fixed) & Project cost (Fixed)
- All financial values must support Versioning (no overwrite)

10. Dashboard & Reports

- Visual analytics (charts & graphs)
- Trend analysis over time
- Decision-making insights
- Admin dashboard — team attendance, leave trends, Billing & Non billing hours.
- Bill hours value, all billing values need to show in USD & INR
- Manager dashboard — team utilization, pending approvals
- Employee dashboard — own attendance, leave balance, logged hours
- Reports — project-wise hours, employee utilization, client-wise billing summary

10.1 Visualization Components

Dashboard must include the following chart types: Revenue Trend (Time Series)

- Type: Line Chart
- Shows:
 - Revenue over time (daily / weekly / monthly)
- Currency:
 - Always displayed in INR (normalized)

10.2. Billable vs Non-Billable Hours

- Type: Bar Chart / Stacked Bar
- Shows:
 - Comparison of billable vs non-billable hours
- Filterable by:

- Date
- Project
- Client

10.3. Revenue by Client

- Type: Pie Chart / Donut Chart
- Shows:
 - Contribution of each client to total revenue

10.4. Project Performance

- Type: Bar Chart
 - Shows:
 - Revenue per project
 - Hours per project
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10.5. Utilization Trend

- Type: Line / Area Chart
- Shows:
 - Total hours logged vs expected hours
 - Helps identify:
 - Under-utilization
 - Over-utilization

10.6 Time-Based Filters (MANDATORY)

All charts must support:

- Date Range:
 - Last 7 days
 - Last 30 days
 - Monthly
 - Custom range

10.7 Drill-Down Capability

- Clicking on any chart element should:
 - Navigate to detailed report
 - Example:
 - Click client → show project-level data

- Click project → show timesheet details

10.8 Data Aggregation Rules

- Only include:
 - Approved timesheets
- Exclude:
 - Draft / Rejected data

10.9 Currency Handling (Charts)

- All charts must:
 - Display values in INR
- Conversion must use:
 - Stored exchange rates (not live recalculation)

10.10 Budget Alert Threshold (%)

Example:

- Default: 70%
- Configurable per client:
 - 60%
 - 70%
 - 80%

Rules:

- Each client can define:
 - Their own **budget consumption alert threshold**
- Applies to:
 - All projects under that client

10.10.1 ALERT LOGIC

Trigger Condition:

If Actual $\geq$ (Configured % $\times$ Budget)

Example:

- Budget = 100 hours
- Threshold = 70%
- Alert triggers at:

- **70 hours**

Applies to:

- T&M → Hours

10.10.2 ALERT CHANNELS

System must trigger alert in:

Dashboard

- Visual warning indicator

Notifications (optional future)

- Project Manager
- Super Admin

10.10.3 DASHBOARD INTEGRATION

Dashboard must show Budget Status Indicators:

-  Green → Below threshold
-  Yellow → Near threshold
-  Red → Exceeded

Metrics:

- Monthly Budget vs Actual
- Consumption %
- Client-level aggregation

10.10.4 DATA AGGREGATION RULE

- Budget tracking must be:
 - Monthly
 - Project-level
 - Aggregated to client level

10.10.5 Performance Requirements

- Dashboard load time:

- ≤ 3 seconds
- Charts rendering:
 - ≤ 2 seconds
- Must use:
 - Cached / pre-aggregated data

10.10.6 Role-Based Dashboard Views

Super Admin (CEO)

- Full dashboard:
 - Revenue
 - Financial metrics
 - All charts

Project Manager

- Limited dashboard:
 - Hours
 - Team utilization
 - Project performance
-  No financial/revenue visibility

Employee

- Personal dashboard:
 - Logged hours
 - Assigned projects
 - Leave status

10.10.7 Export (Integration with Reporting)

- Charts should support:
 - Export as PDF (with visuals)
 - Export data to Excel

10.10.8 Future Enhancements (Optional)

- Forecasting (AI-based revenue prediction)
- Budget vs actual comparison
- Goal tracking (targets vs achieved)

11. Technical Requirements

- Web application (responsive for desktop & tablet)
- Mobile-friendly for timesheet & attendance logging
- Clean, simple & minimalist design
- Secure role-based access control
- Data encryption at rest and in transit
- Automated daily backups
- Hosted on cloud (self-hosted based on preference)
- Daily Currency Price update API Interrogation

12. Tech Stack (Recommended)

- Frontend: **React + Vite**
- UI: **Tailwind CSS**
- Language: **TypeScript**
- Backend: **Node.js + Fastify**
- ORM: **Prisma**
- Database: **MySQL**
- Authentication: **Session-based**
- Scheduler: **System Cron**
- Process Manager: **PM2**
- Web Server: **Nginx**
- AI: **Google Gemini API**
- Architecture: **Modular Monolith**